

Lab: Configuring Firewalls in Microsoft Windows

Estimated time needed: 20 minutes

Introduction

After you complete this hands-on lab, you will be able to create an outbound rule within a Microsoft Windows Firewall. You will know how to block specific IP addresses, using the example of blocking the IP address of the website [Jharkhand State Cricket Association](https://www.cricjarkhand.org/) also seen as <https://www.cricjarkhand.org/>. Understanding how to configure outbound rules in Windows Firewall is essential for managing and securing your network traffic effectively.

Note: If trying these lab instructions on your own computer, your screens might look slightly different than what you see here. Also, to remember to disable or remove any rules you create after completing the lab.

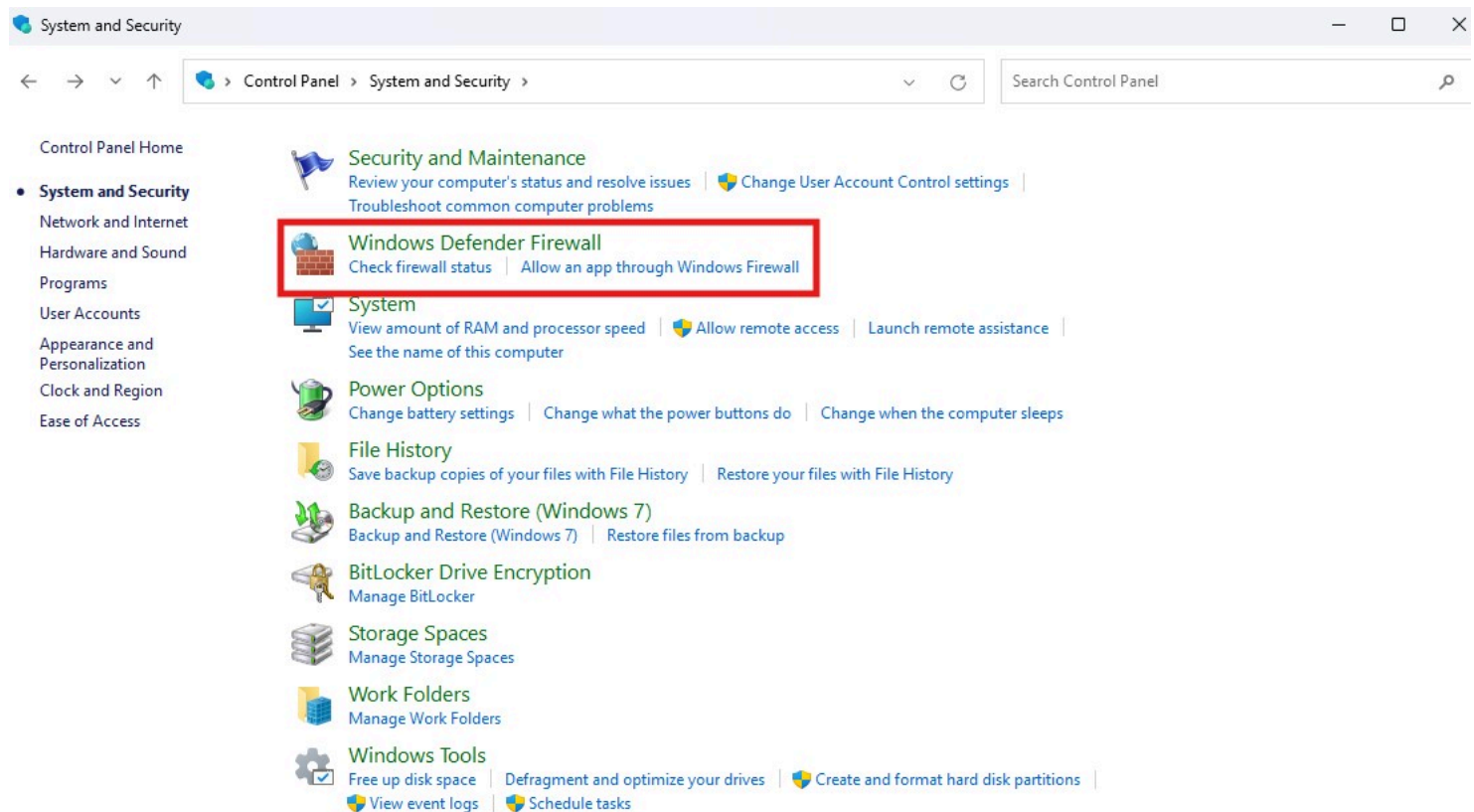
Objectives

By the end of this session, you will be able to:

- Open and navigate the Windows Firewall interface
- Create a new outbound rule in Windows Firewall
- Configure custom settings for outbound rules
- Block specific IP addresses using outbound rules
- Verify the functionality of the newly created rule
- Enable and disable firewall rules

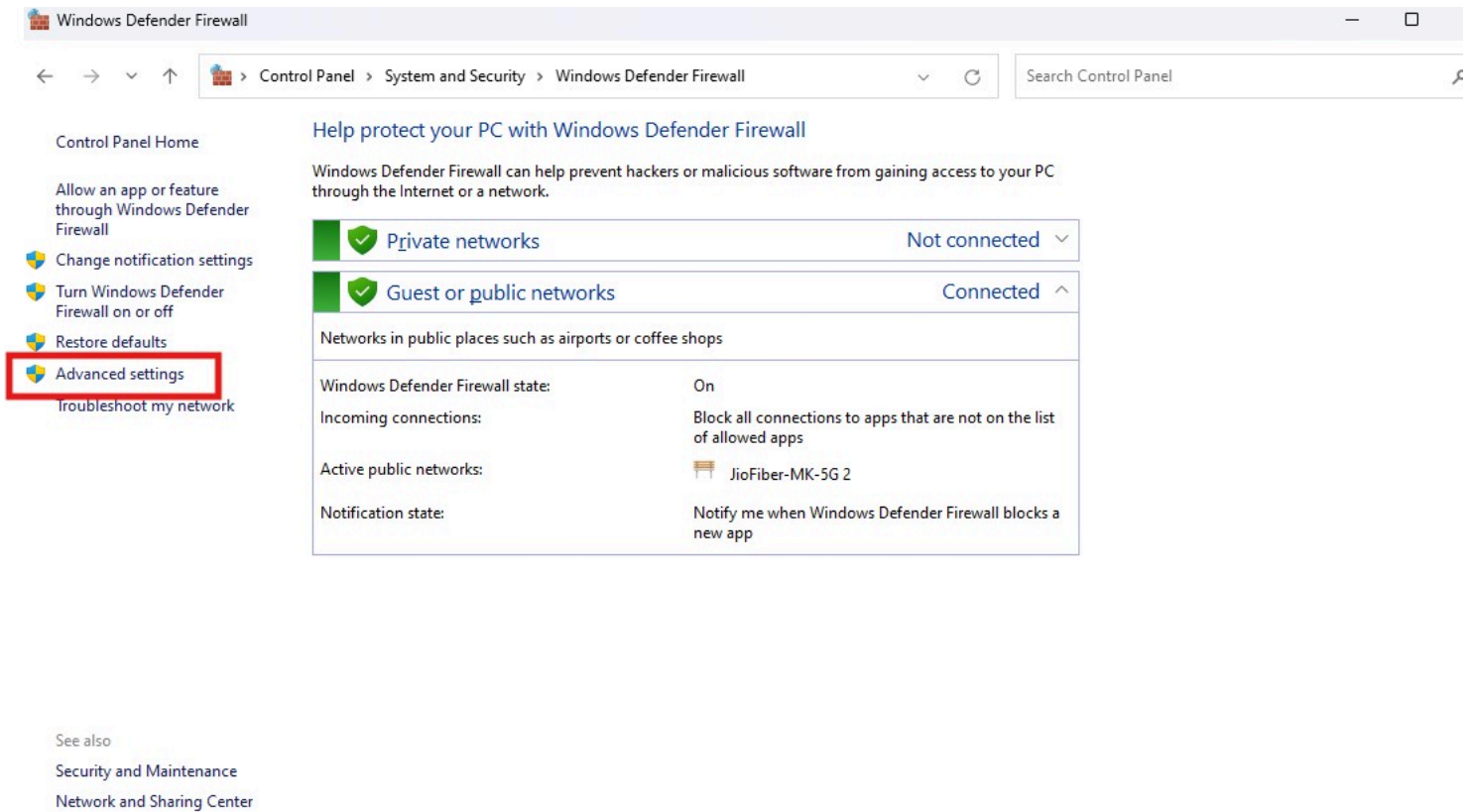
Task 1: Open and navigate the Windows Firewall interface

1. To open Windows Firewall: Press **Win + R** or type **Control panel**, and press **Enter**.
2. Select **System and Security** > **Windows Defender Firewall**.

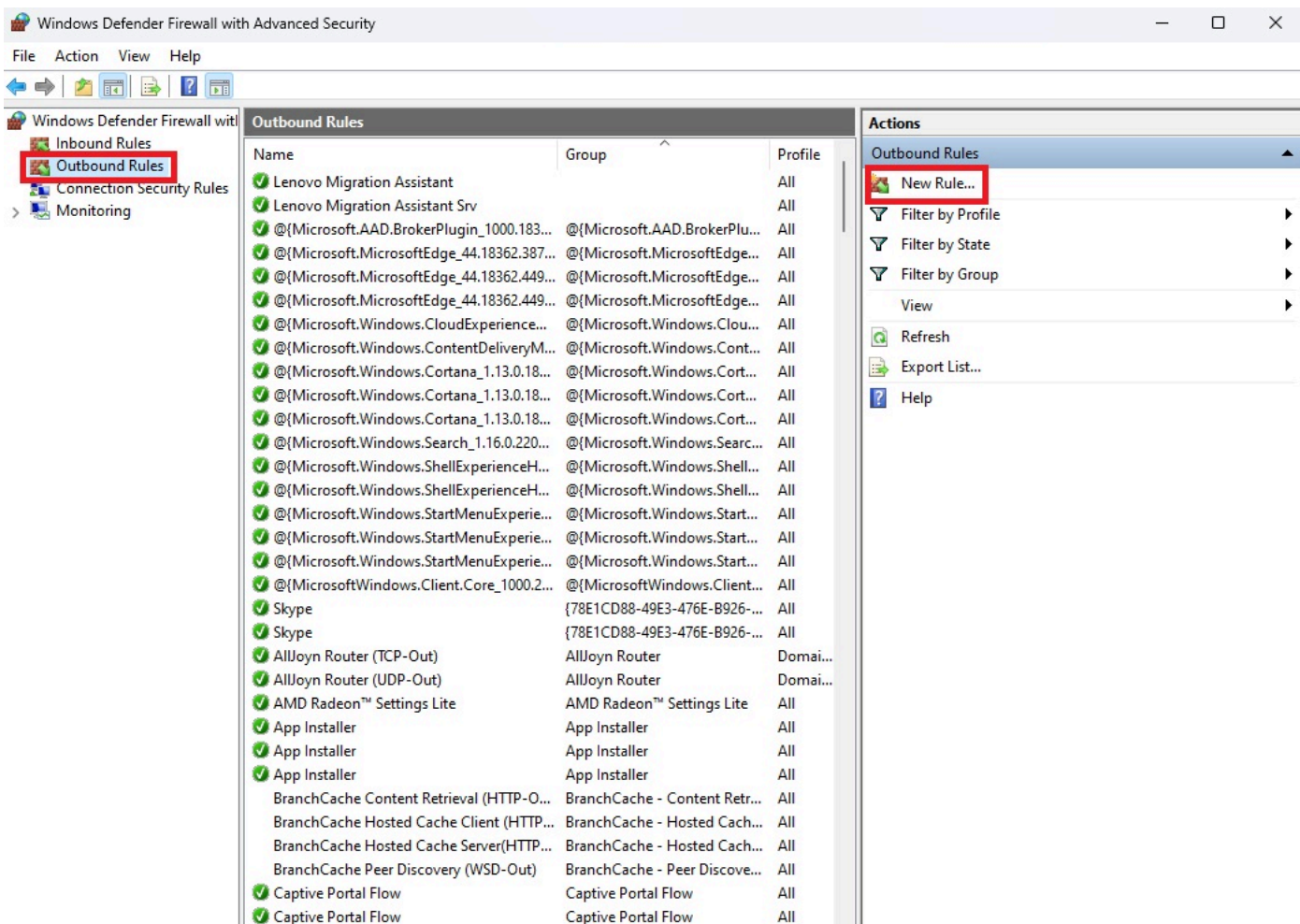


Task 2: Create a new outbound rule in Windows Firewall

1. In the left pane, select **Advanced settings**.

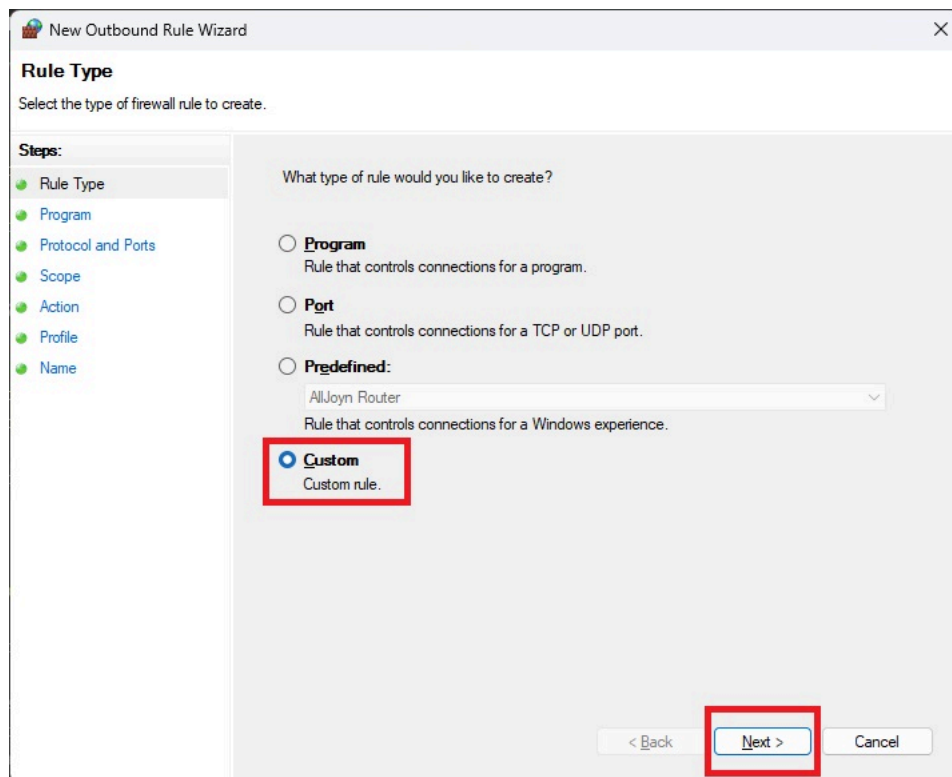


2. In the left pane, select **Outbound Rules**.
3. In the right pane, select **New Rule**.



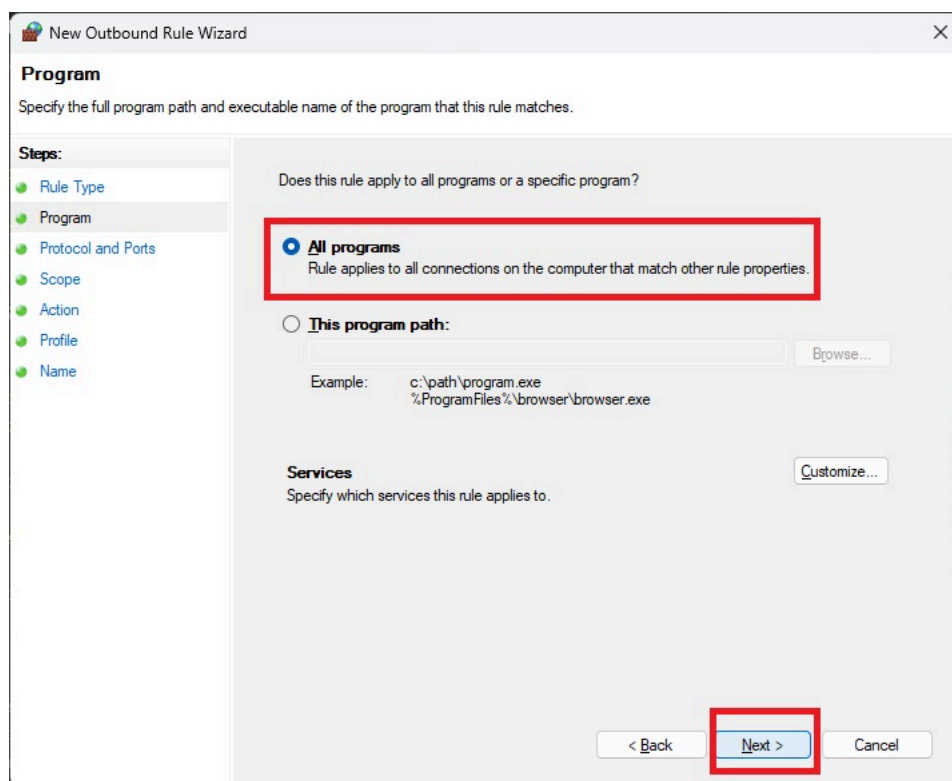
Task 3: Configure custom settings

1. Select **Custom** and **Next**.



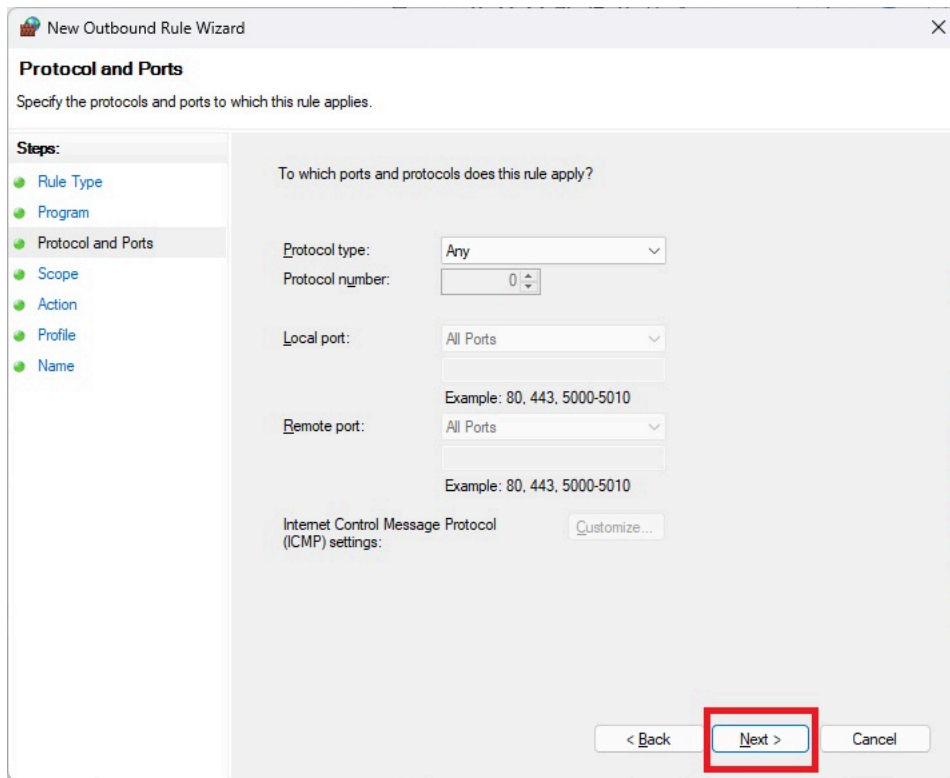
The screenshot shows the 'New Outbound Rule Wizard' window, specifically the 'Rule Type' step. The left sidebar lists the steps: Rule Type, Program, Protocol and Ports, Scope, Action, Profile, and Name. The main area asks 'What type of rule would you like to create?' with three radio button options: **Program** (Rule that controls connections for a program.), **Port** (Rule that controls connections for a TCP or UDP port.), and **Predefined:** (Rule that controls connections for a Windows experience.). The 'Predefined' option has a dropdown menu showing 'AllJoyn Router'. The **Custom** option is selected and highlighted with a red rectangle. At the bottom right, the 'Next >' button is also highlighted with a red rectangle.

2. Select **All programs** and **Next**.



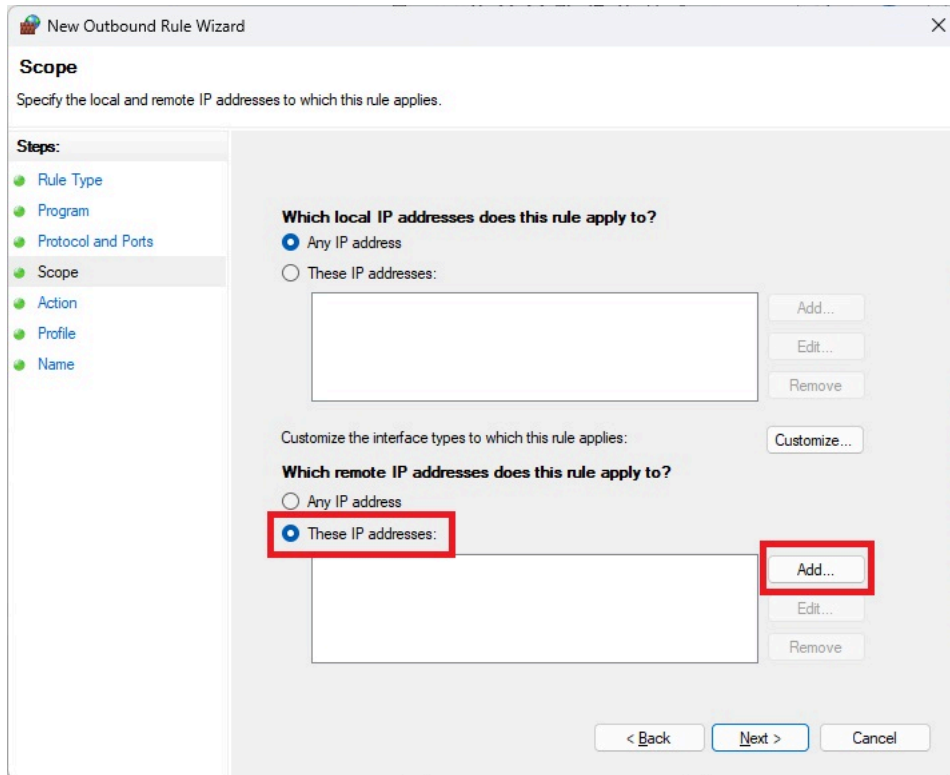
The screenshot shows the 'New Outbound Rule Wizard' window, specifically the 'Program' step. The left sidebar lists the steps: Rule Type, Program, Protocol and Ports, Scope, Action, Profile, and Name. The main area asks 'Does this rule apply to all programs or a specific program?' with two radio button options: **All programs** (Rule applies to all connections on the computer that match other rule properties.) and **This program path:** (with a text box and a 'Browse...' button). The 'All programs' option is selected and highlighted with a red rectangle. Below the 'This program path' section, there is an 'Example:' showing 'c:\path\program.exe' and '%ProgramFiles%\browser\browser.exe'. At the bottom right, the 'Next >' button is highlighted with a red rectangle.

3. Keep the default onscreen parameters and select **Next**.



The screenshot shows the 'New Outbound Rule Wizard' window, specifically the 'Protocol and Ports' step. The left sidebar lists the steps: Rule Type, Program, Protocol and Ports (selected), Scope, Action, Profile, and Name. The main area is titled 'To which ports and protocols does this rule apply?'. It contains the following fields: 'Protocol type' set to 'Any', 'Protocol number' set to '0', 'Local port' set to 'All Ports', and 'Remote port' set to 'All Ports'. Below these, there is an example text 'Example: 80, 443, 5000-5010'. At the bottom, there is a section for 'Internet Control Message Protocol (ICMP) settings' with a 'Customize...' button. The 'Next >' button at the bottom right is highlighted with a red rectangle.

4. Select **These IP Addressess** and **Add**. The **IP Address** dialog box is displayed.



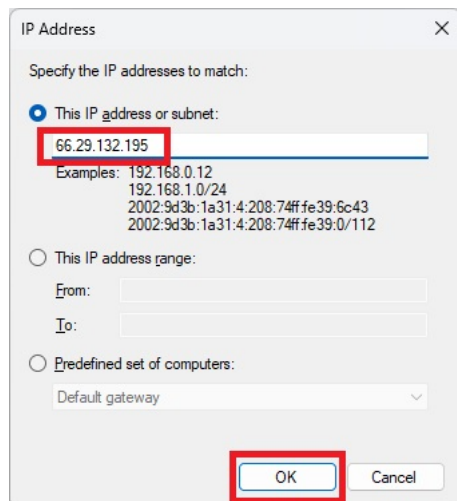
The screenshot shows the 'New Outbound Rule Wizard' window, specifically the 'Scope' step. The left sidebar lists the steps: Rule Type, Program, Protocol and Ports, Scope (selected), Action, Profile, and Name. The main area is titled 'Specify the local and remote IP addresses to which this rule applies.'. It contains two sections: 'Which local IP addresses does this rule apply to?' and 'Which remote IP addresses does this rule apply to?'. In the local section, the 'Any IP address' radio button is selected. In the remote section, the 'These IP addresses:' radio button is selected and highlighted with a red rectangle. Below the remote section, there is a text field for IP addresses and an 'Add...' button, which is also highlighted with a red rectangle. The 'Next >' button at the bottom right is highlighted with a blue border.

Task 4: Block specific IP addresses using outbound rules.

1. Next, you will block the IP Address of 66.29.132.195, which is the address of the website, [Jharkhand State Cricket Association](#) website. Select **This IP address or subnet:** and in the text field type the following the IP address

66.29.132.195

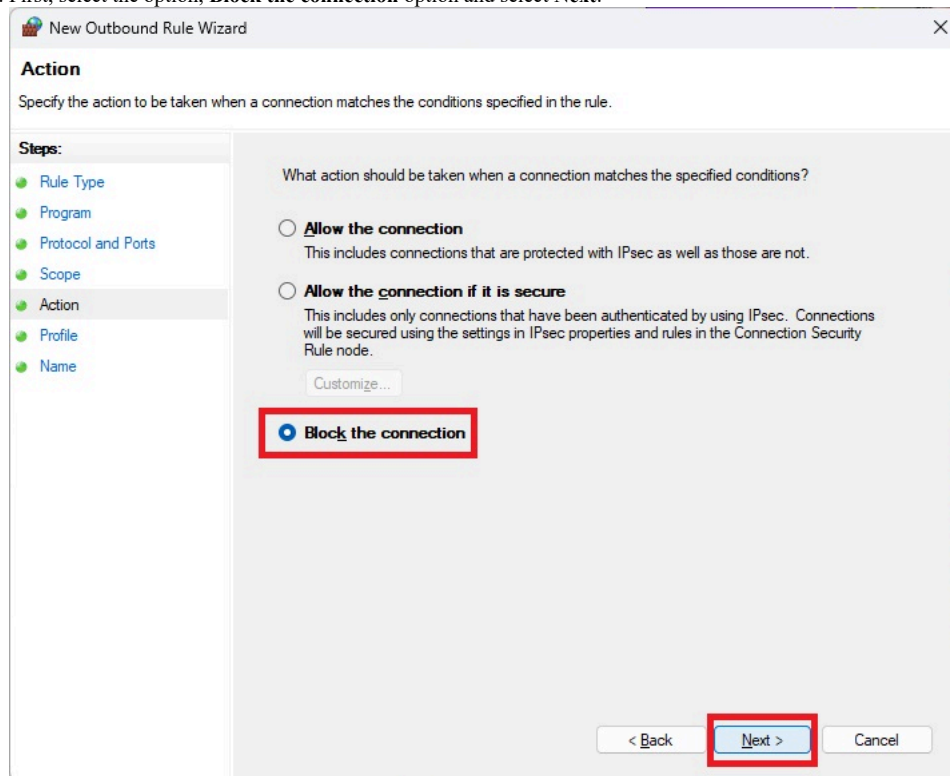
2. Select **OK**. The dialog box closes.



The IP Address dialog box is shown. It has a title bar with a close button. The main text says "Specify the IP addresses to match:". There are three radio buttons: "This IP address or subnet:" (selected), "This IP address range:", and "Predefined set of computers:". The first option has a text box containing "66.29.132.195". Below it are examples of IP addresses and subnets. The second option has "From:" and "To:" text boxes. The third option has a dropdown menu with "Default gateway" selected. At the bottom right are "OK" and "Cancel" buttons.

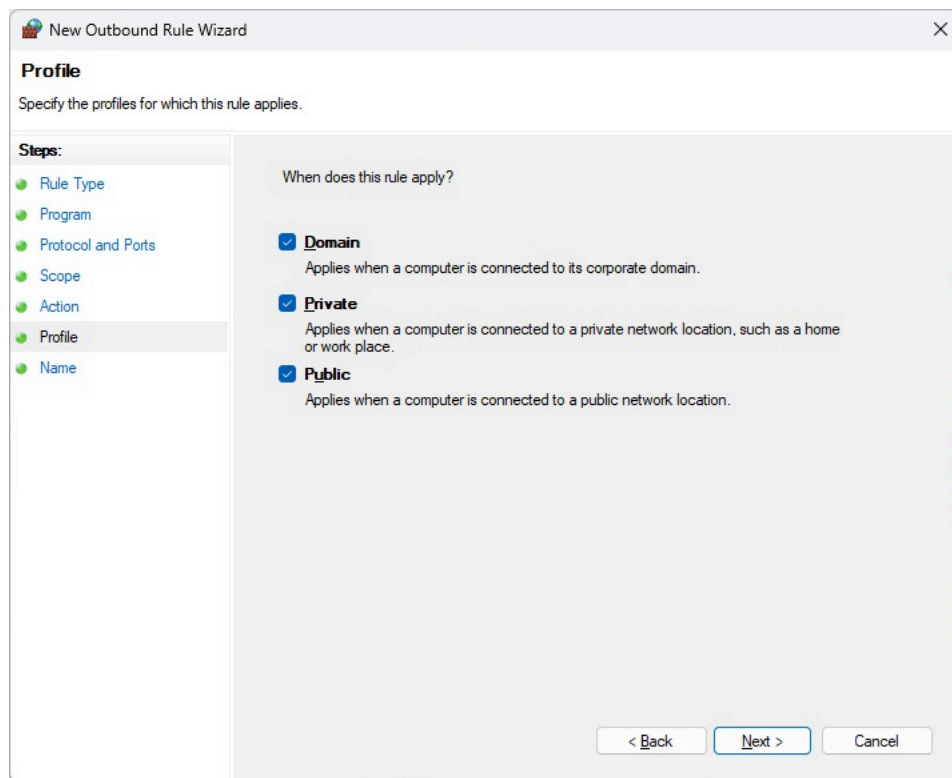
3. The **New Outbound Rule Wizard** window is again displayed.

4. First, select the option, **Block the connection** option and select **Next**.



The New Outbound Rule Wizard window is shown, specifically the "Action" step. The title bar says "New Outbound Rule Wizard". The main text says "Specify the action to be taken when a connection matches the conditions specified in the rule." On the left is a "Steps:" list with "Rule Type", "Program", "Protocol and Ports", "Scope", "Action" (selected), "Profile", and "Name". The main area asks "What action should be taken when a connection matches the specified conditions?". There are two radio buttons: "Allow the connection" and "Block the connection" (selected). Below "Allow the connection" is a description: "This includes connections that are protected with IPsec as well as those are not." Below "Block the connection" is a description: "This includes only connections that have been authenticated by using IPsec. Connections will be secured using the settings in IPsec properties and rules in the Connection Security Rule node." There is a "Customize..." button. At the bottom right are "< Back", "Next >" (selected), and "Cancel" buttons.

5. Select all of the available options for **When does this rule apply?** including **Domain**, **Private**, and **Public**, and select **Next**.



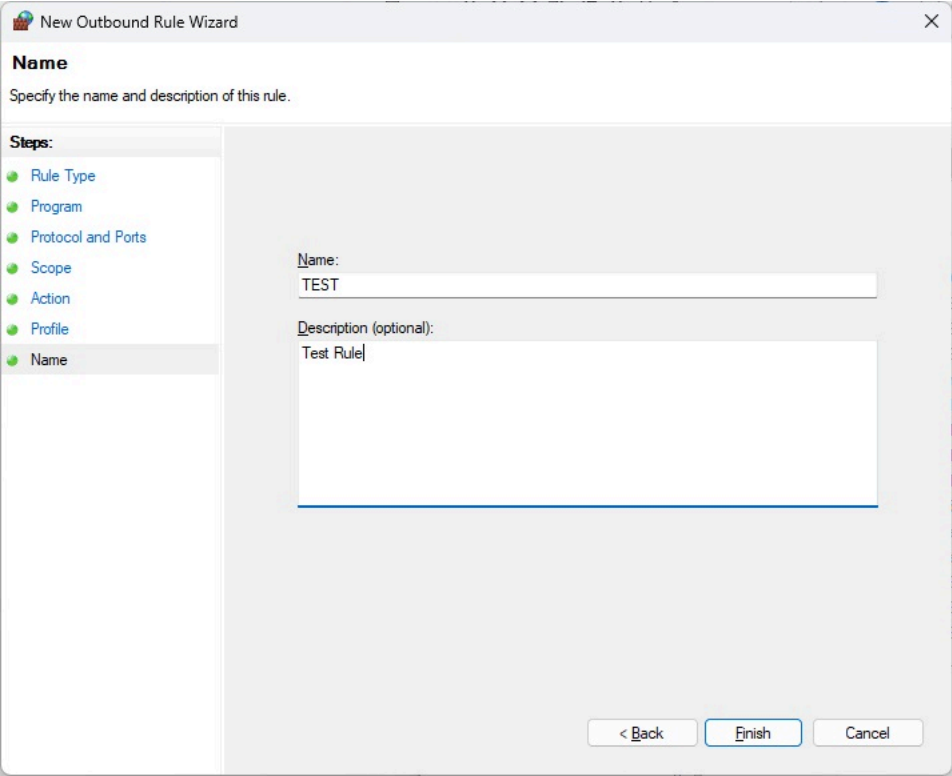
6. Name the newly created rule:

TEST

7. In the description field enter the following text:

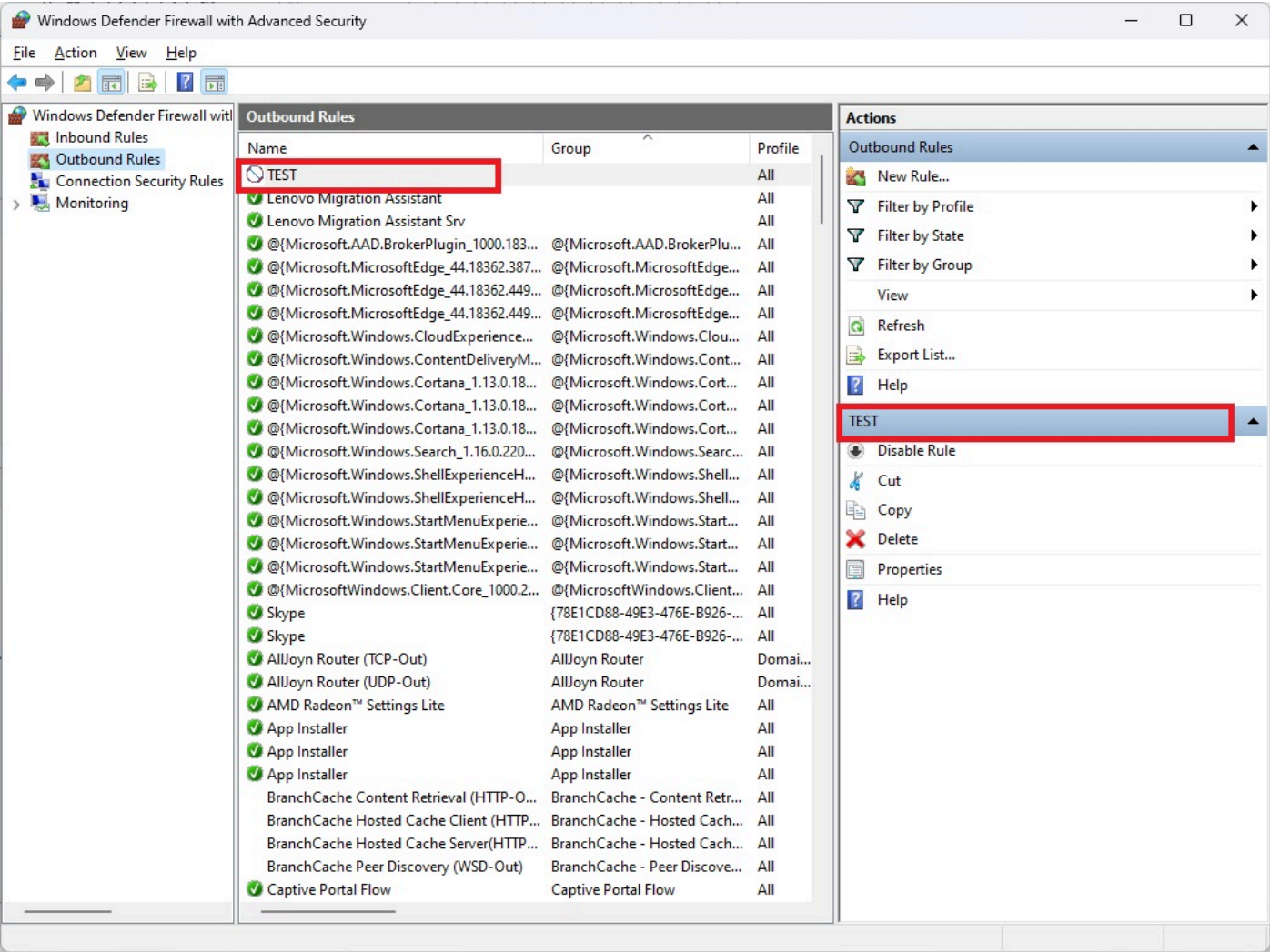
Test rule

8. Select **Finish**.



Task 5: Verify the functionality of the newly created rule

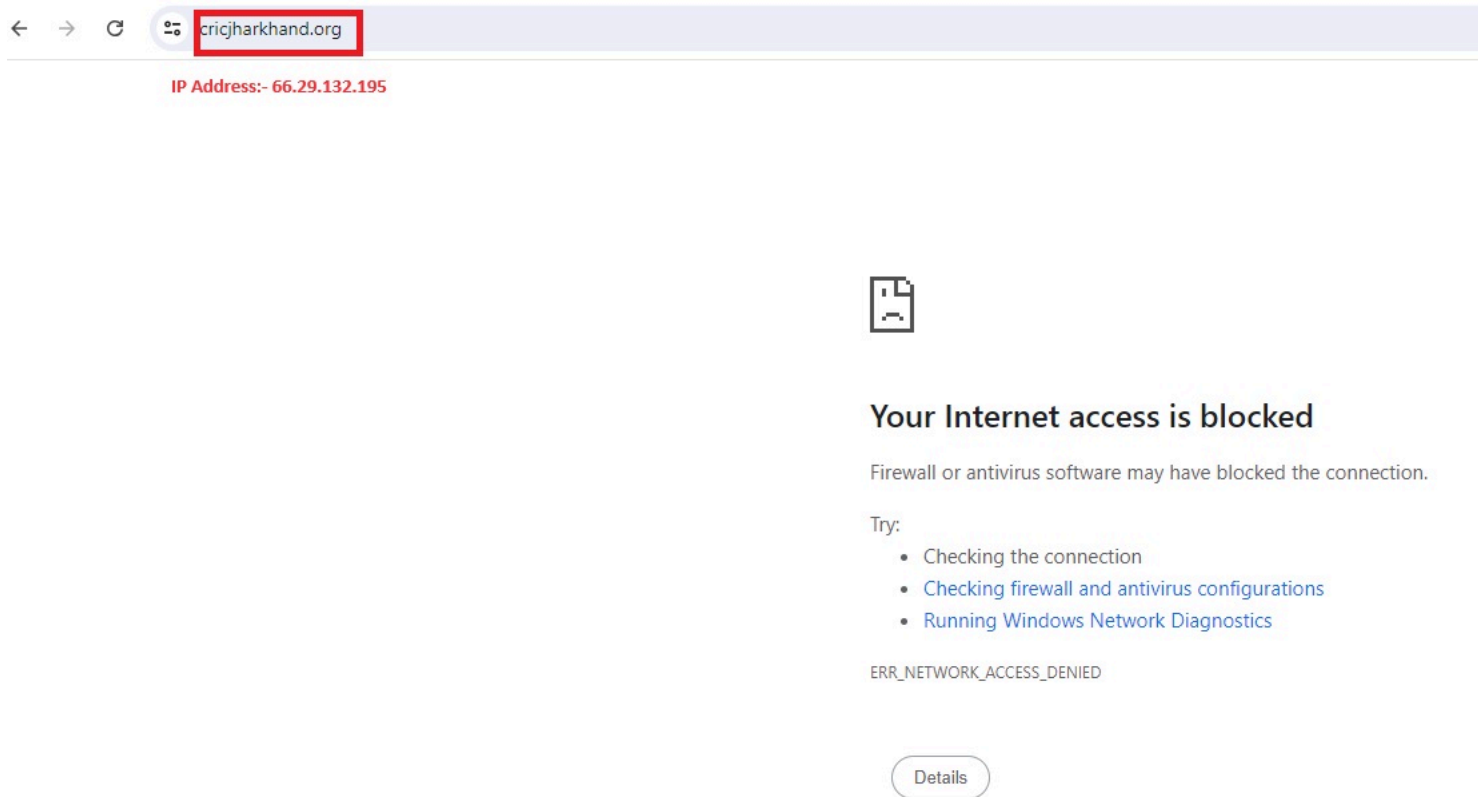
1. If needed, again open the Windows Defender Firewall Advanced Security window.



2. To verify if the newly created rule is working, open a browser window and type or paste the following website in the address bar.

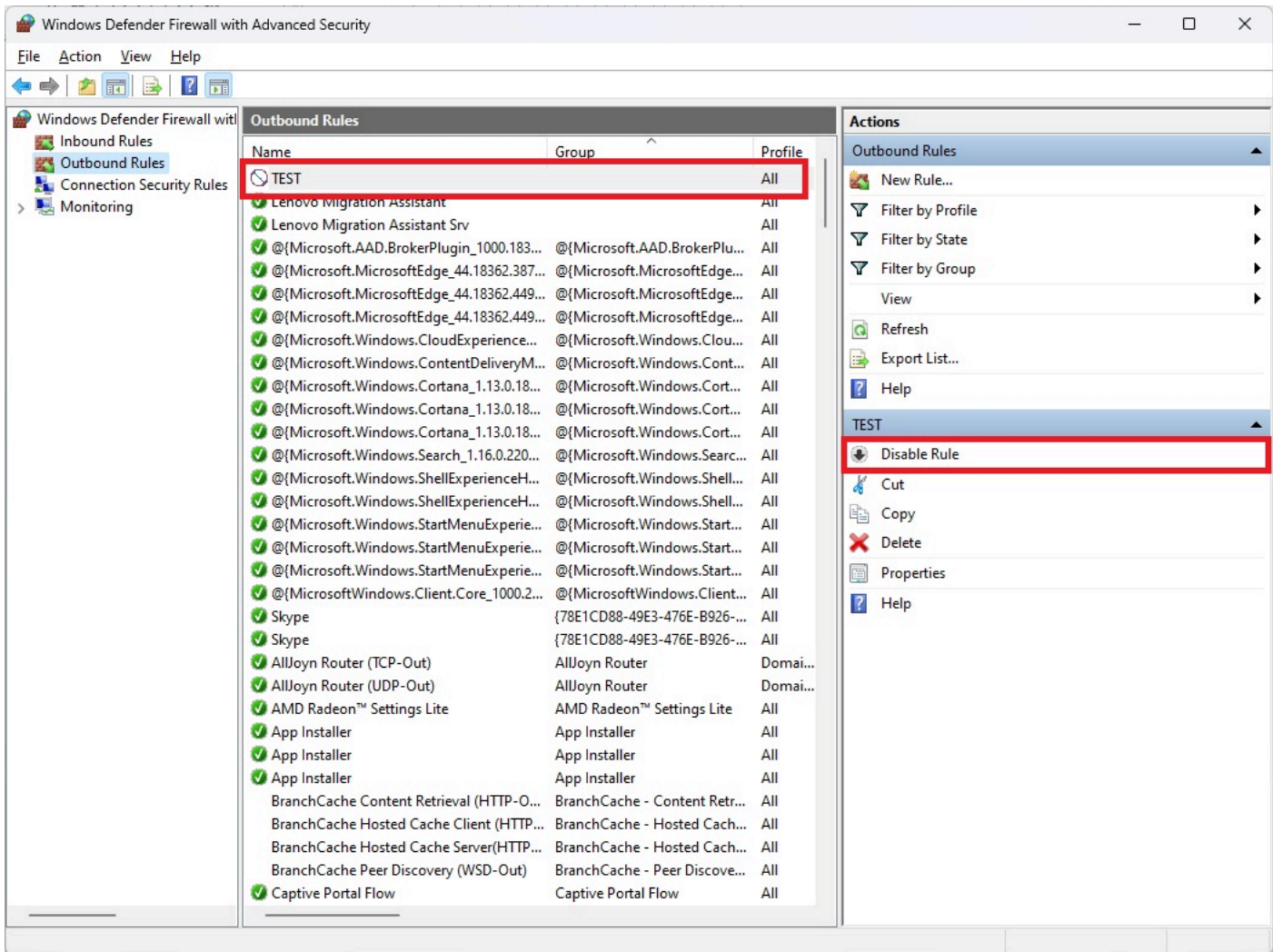
www.cricjharkhand.org

3. You will see a message that indicates that the www.cricjharkhand.org website is blocked or unavailable.



Task 6: Disable the rule to remove the website access restriction

1. Reopen the **Windows Defender Firewall with Advanced Security** screen.
2. Hover over the **TEST** rule you created. Select the option to either **Disable Rule** or to **Delete** the rule.



3. Open a browser and again type or copy the following URL website into the browser address bar.

www.cricjharkhand.org

4. To verify if the newly created rule is working, open a browser window and type, or copy and paste the following website in the address bar.

www.cricjharkhand.org

5. Now, view the website.

Important! If trying this rule on your own computer, remember to disable or remove the rule after completing this exercise.

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Practice Exercises

Next, gain experience by trying these skills on your own.

Exercise 1: Block inbound traffic on a specific port

Problem Statement: Block all inbound traffic on port 445 (used for SMB file sharing) to prevent potential vulnerabilities related to file sharing services.

Hint: You will need to create an inbound rule in Windows Firewall to block traffic on the specified port. Use the inbound port instructions from the lab.

► [Click here for the solution](#)

Exercise 2: Allow outbound traffic only for ppecific programs

Problem Statement: Allow outbound traffic only for a specific browser application, such as Google Chrome, and block all other outbound traffic.

Hint: You will need to create an *outbound* rule to allow traffic for the specific program and block all other outbound traffic.

► [Click here for the solution](#)

Exercise 3: Create a rule to allow remote desktop access

Problem Statement: Create a rule to allow inbound Remote Desktop Protocol (RDP) traffic only from a specific IP address, such as 192.168.1.100.

Hint: You will need to create an *inbound* rule that allows traffic on the RDP port (3389) from a specific IP address.

► [Click here for the solution](#)

Conclusion

You can use **Control Panel** and **Systems and Security** to access your **Windows Defender Firewall settings**, set custom parameters for both outbound and inbound rules, and verify each rule's effectiveness. You can now create an outbound rule to help control and restricting network traffic, contributing to better security management. You can also create an inbound rule to block potential system threats.

You can now locate the **Disable** function within the **Inbound Rules** and **Outbound Rules** screens and restore access to blocked websites, demonstrating the flexibility and control that Windows Defender Firewall rules provide. You're now equipped with the hands-on experience needed to effectively manage firewall rules, enhancing your ability to secure your network environment.

Author(s)

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